H.T.No: | | | | | | | | | | |

Course Code No: 23BS4T02

VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
II B. Tech II Semester (R23) Regular Examinations, April / May – 2025
PROBABILITY AND STATISTICS

(CSE, IT, AI & ML, CSBS)

Time:3hours **Max. Marks: 70M**

Note:1. The Question Paper consists of two parts (**Part-A** and **Part-B**)

2. Answer all the questions in **Part A**

3. Answer either a or b questions from Question number 2 to 6 in **Part-B**

PART –A

1	I	**The means of two samples of sizes 50 and 100, respectively, are 25 and 36, and the standard deviations are 4 and 7. Obtain the mean and standard deviation of the sample of size 150 obtained by combining the two samples.**	L2	CO1	[2M]
	II	**The following are the scores of two batsmen A and B in the IPL 2011:** Score of A 12 115 6 73 7 19 119 36 84 29 Score of B 47 12 16 42 4 51 37 48 13 0 **Who is the better score getter and who is more consistent?**	L2	CO2	[2M]
	III	Explain the equation of an exponential regression model and provide an example of a situation where it might be used.	L1	CO3	[2M]
	IV	What are the key properties of regression coefficients in multiple linear regression?	L1	CO4	[2M]
	V	What is the formula for calculating the expected value of a random variable?	L1	CO5	[2M]
	VI	Explain what a Probability Mass Function (PMF) is used for.	L1	CO1	[2M]
	VII	A sample of 10 observations has a mean of 25. What is the point estimate of the population mean?	L2	CO2	[2M]
	VIII	If the standard deviation is 12 and the sample size is 36, find the maximum error of estimate at 95% confidence level.	L2	CO3	[2M]
	IX	A sample of 36 students has a mean score of 70. The population standard deviation is 12. Test whether the population mean is 75 at 5% level.	L2	CO4	[2M]
	X	A sample of 10 has a mean of 52 and a standard deviation of 4. Test if the population mean is 50.	L2	CO5	[2M]

PART –B

UNIT – I

2(a) I **The marks of 100 students in an examination are shown in the following table. Calculate the mean and median.** L1 CO3 [5M]

Marks obtained:	1 – 10	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60
No of candidates:	3	16	26	31	16	8

II **Calculate Standard Deviation from the following table:** L2 CO3 [5M]

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70
Frequencies	14	45	86	95	50	10

OR

2(b) The first four moments about the point 28.5 of a sample are 0.294, 7.144, 42.409, and 454.98. Calculate the moments about the mean. Also, determine: (a) Coefficient of skewness (γ_1), (b) Coefficient of kurtosis (β_2), (c) Coefficient of excess of kurtosis (γ_2). L2 CO3 [10M]

UNIT – II

3(a) I **Find the correlation coefficient from the following data:** L2 CO2 [5M]

x:	5	7	10	12	15
y:	21	27	30	40	44

II Given $\sum x = 56$, $\sum y = 40$, $\sum x^2 = 524$, $\sum y^2 = 256$, $\sum xy = 364$, and $n = 8$. Find the regression equation of x on y. L1 CO5 [5M]

OR

3(b) I *The correlation coefficient between two variates x and y is $\rho = 0.60$. If $\sigma_x = 1.50$, $\sigma_y = 2.00$, $\bar{x} = 10$, $\bar{y} = 20$, find the equations of the regression lines (a) y on x (b) x on y.* L2 CO2 [5M]

II By the method of least squares, find the straight line that best fits the following data: L2 CO2 [5M]

X	1	2	3	4	5
Y	14	27	40	55	68

UNIT – III

4(a) I *Three soldiers A, B, and C are hitting a target. A hits the target 3 times in 7 shots. B hits 2 times in 6 shots and C hits 3 times in 5 shots. What is the probability that the target is hit?* L2 CO1 [5M]

II *Show that as $p \to 0$ and $n \to \infty$ such that $np \to \lambda$ a constant, the binomial distribution tends to Poisson distribution.* L3 CO3 [5M]

OR

4(b) I Fit a Poisson distribution to the following data given the number of yeast cells per square for 400 squares: L2 CO3 [5M]

No. of cells per sq. :	0	1	2	3	4	5	6	7	8	9	10
No. of squares :	103	143	98	42	8	4	2	0	0	0	0

II If the density function of a continuous random variable X is given by L3 CO3 [4M]

$$f(x) = ax, \quad 0 \le x \le 1$$
$$= a, \quad 1 \le x \le 2$$
$$= 3a - ax, \quad 2 \le x \le 3$$
$$= 0, \quad \text{elsewhere}$$

(i) Find the value of a. (ii) Find the cumulative density function of X.